Sampling Random Variables from $U(0,1)$

The central computational question is simple: a pseudorandom-number generator gives us numbers that are approximately uniform on the unit interval,

$$u \sim U(0,1),$$

and we would like to use them to sample essentially any desired probability distribution.

1 Discrete random variables: tower sampling

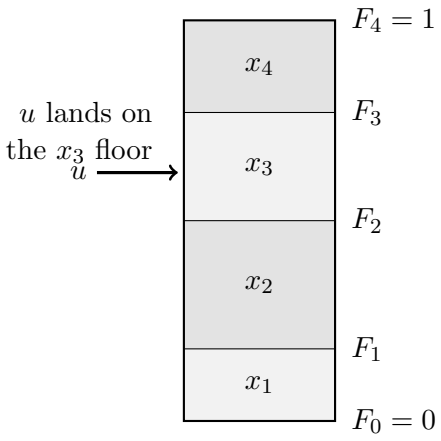
Let a discrete random variable X take values $x_1, \ldots, x_m$ with probability mass function (PMF)

$$\mathbb{P}(X = x_i) = p_i, \qquad p_i \geq 0, \qquad \sum_{i=1}^{m} p_i = 1.$$

Define cumulative probabilities

$$F_k = \sum_{i=1}^{k} p_i, \qquad F_0 = 0.$$

Stack the probabilities vertically to form a *tower* of total height one. Each outcome x_i occupies one *floor* of the tower, whose height is p_i. The boundaries between floors are the cumulative probabilities F_i.



Draw $u \sim U(0,1)$ and imagine dropping it onto the tower at height u. Whichever floor u *lands in* determines the sampled outcome. Algebraically, choose x_k when

$$\boxed{F_{k-1} \leq u < F_k.}$$

The height of floor k is $F_k - F_{k-1} = p_k$. Since a uniform random number lands in an interval with probability equal to its length, x_k is selected with probability p_k. This is the discrete cumulative-probability construction usually called *tower sampling*.

2 Continuous random variables: inverse-transform sampling

For a continuous random variable X with PDF $\pi(x)$, define the cumulative distribution function (CDF)

$$F(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} \pi(x')\,\mathrm{d}x'.$$

If F is invertible, draw $u \sim U(0,1)$ and set

$$\boxed{x = F^{-1}(u).}$$

Indeed,

$$\mathbb{P}(X \le x) = \mathbb{P}\big(F^{-1}(u) \le x\big) = \mathbb{P}\big(u \le F(x)\big) = F(x),$$

because u is uniform on $[0,1]$.

3 Example: waiting time for radioactive decay

Consider a single *unstable radioactive nucleus*. Let T denote its random *decay time* (or *waiting time until decay*). The constant λ is the *decay constant*, with dimensions of inverse time. The defining physical assumption is that, conditional on the nucleus having survived up to the present time, its probability of undergoing radioactive decay during a sufficiently short interval $\mathrm{d}t$ is

$$\mathbb{P}(\text{decay during } \mathrm{d}t \mid \text{nucleus has survived so far}) \simeq \lambda\,\mathrm{d}t.$$

Thus the decay is memoryless: the probability per unit time does not depend on the age of the nucleus. We want the PDF $\pi(t)$ of the random variable T, so that

$$\pi(t)\,\mathrm{d}t \simeq \mathbb{P}(t < T < t + \mathrm{d}t)$$

is the probability that the nucleus survives until time t and then decays during the interval $[t, t + \mathrm{d}t]$.

Divide the preceding time t into n equal intervals of length t/n. During one such interval the probability of decay is approximately $\lambda t/n$, so the probability of survival is

$$1 - \frac{\lambda t}{n}.$$

To decay in $[t, t + \mathrm{d}t]$, the nucleus must survive all n preceding intervals and then decay during the next interval $\mathrm{d}t$. Thus

$$\pi(t)\,\mathrm{d}t = \lim_{n\to\infty} \left(1 - \frac{\lambda t}{n}\right)^{n} \lambda\,\mathrm{d}t.$$

Using

$$\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^{n} = e^{a},$$

we obtain

$$\boxed{\pi(t) = \lambda e^{-\lambda t}, \qquad t \ge 0.}$$

The physics is simply

$$\underbrace{\pi(t)\,\mathrm{d}t}_{\substack{\text{decay between} \\ t \text{ and } t+\mathrm{d}t}} = \underbrace{e^{-\lambda t}}_{\text{survival probability } S(t)} \underbrace{\lambda\,\mathrm{d}t}_{\substack{\text{conditional probability to decay} \\ \text{during the next } \mathrm{d}t}}.$$

The survival probability is therefore

$$S(t) = \mathbb{P}(T > t) = e^{-\lambda t},$$

and the CDF of the decay time is

$$F(t) = \mathbb{P}(T \le t) = 1 - S(t) = 1 - e^{-\lambda t}.$$

Inverse-transform sampling gives

$$u = 1 - e^{-\lambda t} \quad \Longrightarrow \quad t = -\frac{1}{\lambda}\ln(1-u).$$

Since $1-u$ is also uniform on $(0,1)$, in practice one usually writes

$$\boxed{t = -\frac{1}{\lambda}\ln u.}$$

4 Gaussian random variables: Box–Muller

For a standard normal variable,

$$\pi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

the inverse CDF is not an elementary function. The Box–Muller construction instead considers two independent standard normal variables X and Y. Their joint PDF is

$$\pi(x,y) = \frac{1}{2\pi}\exp\left[-\frac{x^2+y^2}{2}\right].$$

The infinitesimal probability is not just the PDF; it is the PDF multiplied by the corresponding volume element:

$$\mathrm{d}\mathbb{P} = \pi(x,y)\,\mathrm{d}x\,\mathrm{d}y.$$

Now transform to polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

A small coordinate element $\mathrm{d}r\,\mathrm{d}\theta$ is not itself an area in the Cartesian plane. Its radial length is $\mathrm{d}r$, while its tangential length is $r\,\mathrm{d}\theta$. Hence the corresponding Cartesian area is

$$\boxed{\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta.}$$

Equivalently,

$$\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| = r.$$

Therefore

$$\mathrm{d}\mathbb{P} = \frac{1}{2\pi}e^{-r^2/2}\,r\,\mathrm{d}r\,\mathrm{d}\theta = \left(\frac{\mathrm{d}\theta}{2\pi}\right)\left(re^{-r^2/2}\,\mathrm{d}r\right).$$

The measure factorizes. Thus

$$\theta \sim U(0,2\pi).$$

For the radial part define

$$s = \frac{r^2}{2}, \qquad \mathrm{d}s = r\,\mathrm{d}r.$$

Then

$$\pi_s(s)\,\mathrm{d}s = e^{-s}\,\mathrm{d}s,$$

so s has the exponential distribution with rate one. From the previous section,

$$s = -\ln u_1, \qquad \theta = 2\pi u_2,$$

where u_1, u_2 are independent $U(0,1)$ variables. Hence

$$r = \sqrt{-2\ln u_1},$$

and transforming back gives

$$\boxed{x = \sqrt{-2\ln u_1}\cos(2\pi u_2),}$$

$$\boxed{y = \sqrt{-2\ln u_1}\sin(2\pi u_2).}$$

Thus two independent uniform random numbers generate two independent standard normal random numbers.

5 The Jacobian as a volume-conversion factor

The preceding calculation illustrates the elementary geometric meaning of a Jacobian. Suppose

$$\boldsymbol{z} = (z_1, \ldots, z_d)$$

is described instead by coordinates

$$\boldsymbol{q} = (q_1, \ldots, q_d).$$

A small coordinate box

$$\mathrm{d}^d q = \mathrm{d}q_1 \cdots \mathrm{d}q_d$$

need not have the same volume when mapped into Cartesian $\boldsymbol{z}$-space. To leading order, the Cartesian volume represented by this coordinate element is

$$\boxed{\mathrm{d}^d z = \left|\det \frac{\partial(z_1, \ldots, z_d)}{\partial(q_1, \ldots, q_d)}\right| \mathrm{d}^d q.}$$

The Jacobian is therefore the local volume-conversion factor. The probability itself is coordinate independent:

$$\boxed{\mathrm{d}\mathbb{P} = \pi_{\boldsymbol{z}}(\boldsymbol{z})\,\mathrm{d}^d z = \pi_{\boldsymbol{q}}(\boldsymbol{q})\,\mathrm{d}^d q.}$$

The PDFs change under a coordinate transformation precisely because the coordinate volume elements change.

6 Uniform points on a d-dimensional sphere

Before deriving the algorithm, it is useful to be precise about notation. A point

$$\boldsymbol{z} = (z_1, \ldots, z_d) \in \mathbb{R}^d$$

has d Cartesian components. Its direction lies on the unit sphere

$$S^{d-1} = \{\boldsymbol{x} \in \mathbb{R}^d : |\boldsymbol{x}| = 1\}.$$

Thus the unit circle is $S^1 \subset \mathbb{R}^2$, the ordinary spherical surface is $S^2 \subset \mathbb{R}^3$, and in general normalizing a vector in $\mathbb{R}^d$ produces a point on S^{d-1}.

The phrase *uniform on the sphere* means uniform with respect to surface area: two surface patches having the same area must have the same probability. It does *not* mean that the angular coordinates themselves should all be sampled uniformly. For example, on the ordinary sphere S^2, choosing both θ and ϕ uniformly produces too many points near the poles because the surface element is

$$\mathrm{d}\Omega = \sin\theta \; \mathrm{d}\theta \, \mathrm{d}\phi.$$

The Gaussian-normalization method below avoids having to work out these angular Jacobians explicitly.

We now use the same measure argument in d dimensions. Generate

$$z_i \sim N(0,1), \qquad i = 1, \ldots, d,$$

independently. Their joint Cartesian PDF is

$$\pi(\boldsymbol{z}) = \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}} e^{-z_i^2/2} = \frac{1}{(2\pi)^{d/2}} e^{-r^2/2},$$

where

$$r = |\boldsymbol{z}| = \sqrt{\sum_{i=1}^{d} z_i^2}.$$

The infinitesimal probability is

$$\mathrm{d}\mathbb{P} = \pi(\boldsymbol{z}) \, \mathrm{d}^d z.$$

In d-dimensional spherical coordinates, let Ω denote all $d-1$ angular coordinates. The Cartesian volume represented by $\mathrm{d}r \, \mathrm{d}\Omega$ is

$$\boxed{\mathrm{d}^d z = r^{d-1} \, \mathrm{d}r \, \mathrm{d}\Omega.}$$

Hence

$$\mathrm{d}\mathbb{P} = \frac{1}{(2\pi)^{d/2}} e^{-r^2/2} r^{d-1} \, \mathrm{d}r \, \mathrm{d}\Omega.$$

Everything multiplying $\mathrm{d}\Omega$ depends only on r, not on direction. Integrating over all radii gives

$$\mathrm{d}\mathbb{P}_{\text{direction}} = C \, \mathrm{d}\Omega,$$

where C is independent of Ω. Normalization over the total solid angle Ω_{d-1} gives

$$\boxed{\mathrm{d}\mathbb{P}_{\text{direction}} = \frac{\mathrm{d}\Omega}{\Omega_{d-1}}.}$$

Thus the direction of a d-dimensional Gaussian vector is uniformly distributed over the sphere. Therefore

$$\boxed{\hat{\boldsymbol{n}} = \frac{\boldsymbol{z}}{|\boldsymbol{z}|}}$$

is uniformly distributed on the unit sphere S^{d-1}.

In practice the algorithm is therefore simply:

1. Generate d independent standard normal numbers $z_1, \ldots, z_d$.

2. Compute $r = \sqrt{z_1^2 + \cdots + z_d^2}$.

3. Return
$$\hat{n}_i = \frac{z_i}{r}, \qquad i = 1, \ldots, d.$$

The Gaussian vector is not uniformly distributed in radius, but this does not matter: its distribution is rotationally invariant, so every direction is equally likely. Dividing by $|\boldsymbol{z}|$ removes the radius and keeps only that uniformly distributed direction.

7 Uniform points inside a d-dimensional ball

Now consider a point uniformly distributed in Cartesian volume inside a ball of radius R. "Uniform" means that the Cartesian PDF is constant, so
$$\mathrm{d}\mathbb{P} \propto \mathrm{d}^d x.$$

Using spherical coordinates,
$$\mathrm{d}^d x = r^{d-1}\,\mathrm{d}r\,\mathrm{d}\Omega,$$

and therefore
$$\mathrm{d}\mathbb{P} \propto r^{d-1}\,\mathrm{d}r\,\mathrm{d}\Omega.$$

Again the angular distribution is uniform, while the radial PDF is
$$\pi_r(r) \propto r^{d-1}.$$

Normalization gives
$$\boxed{\pi_r(r) = \frac{d}{R^d} r^{d-1}, \qquad 0 \le r \le R.}$$

This factor has a direct geometric interpretation. A thin shell between r and $r + \mathrm{d}r$ has volume
$$\mathrm{d}V = \Omega_{d-1} r^{d-1}\,\mathrm{d}r,$$

so a uniformly sampled point must be more likely to fall at larger r simply because there is more Cartesian volume there. Equivalently, the probability that the sampled point lies inside radius r is the ratio of the two ball volumes,
$$F(r) = \frac{V_d(r)}{V_d(R)} = \left(\frac{r}{R}\right)^d.$$

The same result follows by integrating the radial PDF:
$$F(r) = \int_0^r \frac{d}{R^d} (r')^{d-1}\,\mathrm{d}r' = \left(\frac{r}{R}\right)^d.$$

Applying inverse-transform sampling,
$$u = \left(\frac{r}{R}\right)^d \quad \Longrightarrow \quad \boxed{r = R u^{1/d}.}$$

Thus a uniform point in the ball is generated by
$$z_i \sim N(0,1), \qquad \hat{\boldsymbol{n}} = \frac{\boldsymbol{z}}{|\boldsymbol{z}|}, \qquad u \sim U(0,1),$$

followed by

$$\boxed{\boldsymbol{x} = Ru^{1/d}\hat{\boldsymbol{n}}.}$$

Choosing r uniformly from $[0, R]$ would be incorrect for $d > 1$: it would overweight the center. The factor r^{d-1} is the increasing amount of Cartesian volume available in a shell at radius r.

So there are two separate ingredients:

1. sample a *direction* uniformly on S^{d-1} using a normalized Gaussian vector;
2. sample the *radius* from $\pi_r(r)$, which is accomplished by $r = Ru^{1/d}$.

These two random choices are independent. In pseudocode,

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for i = 1,...,d:
    z[i] = Gaussian(0,1)
n = z / norm(z)
u = Uniform(0,1)
r = R * u^(1/d)
x = r * n
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A common mistake is to generate a uniform direction and then choose $r \sim U(0, R)$. That produces a distribution that is uniform in radius, not uniform in Cartesian volume.

8 Conceptual chain

The methods form a connected toolkit:

$\boxed{U(0,1) \longrightarrow \text{tower sampling} \longrightarrow \text{inverse transform} \longrightarrow \text{exponential waiting time} \longrightarrow \text{Box–Muller} \longrightarrow \text{Gaussian vecto}}$

Two ideas recur throughout: transform a difficult distribution into one that is easy to sample, and keep track of the full probability measure—PDF π times the appropriate volume element.